

devices from Nexperia. The two protection devices are optimised for use with retimers and redrivers on high-speed data lines.



PESD2V8Y1BSF is capable of protecting USB4 (Thunderbolt)

interfaces while PESD4V0Y1BCSF is capable of protecting the HDMI 2.1 interface in addition to the USB4. The RF performance of the TrEOS technology used by the devices remains stable over the full operating voltage range. Apart from USB4 and HDMI 2.1, both can also be used to protect PCIe and DisplayPort interfaces.

Nexperia

<https://www.nexperia.com>

Compact 40V MOSFET

Diodes Incorporated's DMTH4M-70SPGWQ is a 40V MOSFET that features a low RDS(ON) of 0.54mΩ at a gate drive of 10V, while its gate charge



is 117nC. The product helps in achieving high system efficiency in

applications such as high-power BLDC motor drives, DC-DC converters and charging systems, while ensuring low power dissipation to achieve high efficiency. It is available in a 64mm² package, which occupies 40% less space than the TO263 (D2PAK) package format.

Diodes Incorporated

<https://www.diodes.com>

Processor for small-size HMI

Sitara AM62 processor can enable support for small-size human-machine

interface (HMI) applications and dual-screen displays. The low-cost solution



New Sitara processors with edge AI capabilities reduce power consumption by up to 50%

can enable basic camera based image processing and edge AI func-

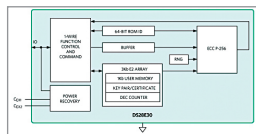
tion, which includes pattern recognition and image detection. It can also allow gesture recognition in a noisy environment and use wireless connectivity to enable control of machines by tablets or phones.

Texas Instruments

<https://www.ti.com>

Secure authenticator

Analog Devices DS28E30 is an ECDSA secure authenticator to protect products from counterfeiting. It combines



ECDSA along with applications data and secure stor-

age of keys. The cost-effective solution comes in a compact package and requires only a single wire to operate, making it easy to integrate into new or existing products. The secure authenticator can be employed in a wide range of applications, such as asset tracking, battery ID authentication, etc.

Analog Devices

<https://www.analog.com/en/index.html>

Gas-impregnated PFC capacitor



TDK Corporation's B25674C is an EPCOS gas-impregnated PFC capacitor series that covers a voltage range from 230V to 690V AC. The series can be used at hot-

spot temperatures up to 85°C and have a long lifetime of up to 240,000 hours in temperature class -40/D.

The capacitors can handle high inrush currents equivalent to 500 times the rated current.

TDK Corporation

<https://www.tdk.com/en/index.html>

EQUIPMENT

Precise time scale system

Microchip's fully integrated precise time scale system (PTSS) can act as a backup and enables nations to operate



independent time scale back-up to GNSS to protect critical

infrastructure. The PTSS is a world-class timing system that is traceable to Universal Coordinated Time (UTC) and does not depend on GNSS. It is a fully integrated system capable of providing timing accuracies comparable to the world's best national laboratories.

Microchip Technologies

<https://www.microchip.com>

MODULES AND BOARDS

Automotive DC-DC converters

The RPX series of DC/DC converters from RECOM RPX now meets the AEC-Q100 Grade 1 requirement, making it



more secure and reliable for automotive applications. The new converters come in a QFN package and include



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